

Rain in the rainforest

An equatorial rainforest gets rain every day (since it is a rainforest) and does not have much seasonal weather variation (since it is equatorial). We model the daily rainfall as independent random variables with exponential(λ) distributions for some fixed λ . Unlike our previous examples, these exponential random variables are not waiting times.

1. Suppose that the average annual rainfall in the forest is 80 inches. What value would you choose for λ ? (Use the “method of moments”: match the expected daily rainfall with the average daily rainfall.)
 2. According to your model, what is the probability that there will be more than 1 inch of rain on any given day?
 3. Suppose that you want to model the number of days in a year with more than 1 inch of rain using a random variable N . What distribution would you use for N ?
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4. According to your random variable N , how many days in a year would you expect with more than 1 inch of rain? What is the standard deviation in the number of days with more than 1 inch of rain?

Expected number of days in a year with more than 1 inch of rain:

Standard deviation with units of measurement:

5. Suppose that it rained more than 1 inch today in the rainforest. Let T be the random variable (for our model) counting the number of days to wait for the next day with more than 1 inch of rain. What distribution would you use for T ?

T

